

LAPORAN MODUL PRAKTIKUM PEMROGRAMAN SISI BACK END



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**PROGRAM STUDI INFORMATIKA
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MODUL 9

PHP DATABASE (Mengisi Tabel)

A. TUJUAN PRAKTIKUM

1. Memahami cara koneksi dan akses data ke tabel

B. PEMBAHASAN LISTING PRAKTIK

PRAKTIK

Praktik 1

```
MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS storedb;
Query OK, 0 rows affected, 1 warning (0.001 sec)

MariaDB [(none)]> CREATE USER 'store'@'localhost' IDENTIFIED BY 'store';
Query OK, 0 rows affected (0.007 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON storedb.* TO 'store'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]>
```

Pembahasa : Program praktik 1 adalah perintah SQL untuk membuat database dan user. Perintah `CREATE DATABASE IF NOT EXISTS storedb` membuat database bernama `storedb`, kemudian `CREATE USER 'store'@'localhost' IDENTIFIED BY 'store'` membuat user baru bernama `store` dengan password `store` yang hanya bisa mengakses dari `localhost`. Perintah `GRANT ALL PRIVILEGES ON storedb.* TO 'store'@'localhost'` memberikan seluruh hak akses user `store` terhadap semua tabel di database `storedb`, dan `FLUSH PRIVILEGES` memastikan perubahan langsung diterapkan oleh server.

Praktik 2

```

MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS storedb;
Query OK, 1 row affected (0.007 sec)

MariaDB [(none)]> CREATE USER 'store'@'localhost' IDENTIFIED BY 'store';
ERROR 1396 (HY000): Operation CREATE USER failed for 'store'@'localhost'
MariaDB [(none)]> GRANT ALL PRIVILEGES ON storedb.* TO 'store'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]>

```

```

MariaDB [storedb]> SELECT * FROM products;
+-----+-----+-----+-----+-----+-----+-----+
| code | name      | description      | price | discount | status_code | category_code |
+-----+-----+-----+-----+-----+-----+-----+
| 1    | Pensil 2B | Pensil berkualitas | 5000  | 10       | 1           | 1             |
| 2    | Laptop ASUS | Laptop gaming    | 8000000 | 5        | 1           | 2             |
| 3    | Kaos Polos | Kaos cotton      | 50000  | 15       | 1           | 3             |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.007 sec)

MariaDB [storedb]>

```

```

MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS storedb;
Query OK, 1 row affected (0.007 sec)

MariaDB [(none)]> CREATE USER 'store'@'localhost' IDENTIFIED BY 'store';
ERROR 1396 (HY000): Operation CREATE USER failed for 'store'@'localhost'
MariaDB [(none)]> GRANT ALL PRIVILEGES ON storedb.* TO 'store'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

```

```

MariaDB [(none)]> USE storedb;
Database changed
MariaDB [storedb]>
MariaDB [storedb]> CREATE TABLE IF NOT EXISTS category (
->     code INT(5) NOT NULL AUTO_INCREMENT,
->     category VARCHAR(25),
->     PRIMARY KEY (code)
-> );
Query OK, 0 rows affected (0.040 sec)

MariaDB [storedb]>
MariaDB [storedb]> CREATE TABLE IF NOT EXISTS products (
->     code INT(5) NOT NULL AUTO_INCREMENT,
->     name VARCHAR(35),
->     description VARCHAR(35),
->     price INT(8),
->     discount INT(2),
->     status_code INT(1),
->     category_code INT(1),
->     PRIMARY KEY (code)
-> );
Query OK, 0 rows affected (0.011 sec)

```

```

MariaDB [storedb]>
MariaDB [storedb]> INSERT INTO category (category) VALUES ('Alat Tulis');
Query OK, 1 row affected (0.003 sec)

MariaDB [storedb]> INSERT INTO category (category) VALUES ('Elektronik');
Query OK, 1 row affected (0.002 sec)

MariaDB [storedb]> INSERT INTO category (category) VALUES ('Pakaian');
Query OK, 1 row affected (0.002 sec)

MariaDB [storedb]>
MariaDB [storedb]> INSERT INTO products (name, description, price, discount, status_code, category_code) VALUES ('Pensil 2B', 'Pensil berkualitas', 5000, 10, 1, 1);
Query OK, 1 row affected (0.003 sec)

MariaDB [storedb]> INSERT INTO products (name, description, price, discount, status_code, category_code) VALUES ('Laptop ASUS', 'Laptop gaming', 8000000, 5, 1, 2);
Query OK, 1 row affected (0.002 sec)

MariaDB [storedb]> INSERT INTO products (name, description, price, discount, status_code, category_code) VALUES ('Kaos Polos', 'Kaos cotton', 50000, 15, 1, 3);
Query OK, 1 row affected (0.016 sec)

MariaDB [storedb]> SHOW TABLES;
+-----+
| Tables_in_storedb |
+-----+
| category           |
| products           |
+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT * FROM products;
+----+-----+-----+-----+-----+-----+-----+
| code | name      | description | price | discount | status_code | category_code |
+----+-----+-----+-----+-----+-----+-----+
| 1    | Pensil 2B | Pensil berkualitas | 5000 | 10      | 1           | 1             |
| 2    | Laptop ASUS | Laptop gaming | 8000000 | 5      | 1           | 2             |
| 3    | Kaos Polos | Kaos cotton | 50000 | 15      | 1           | 3             |
+----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)

```

```

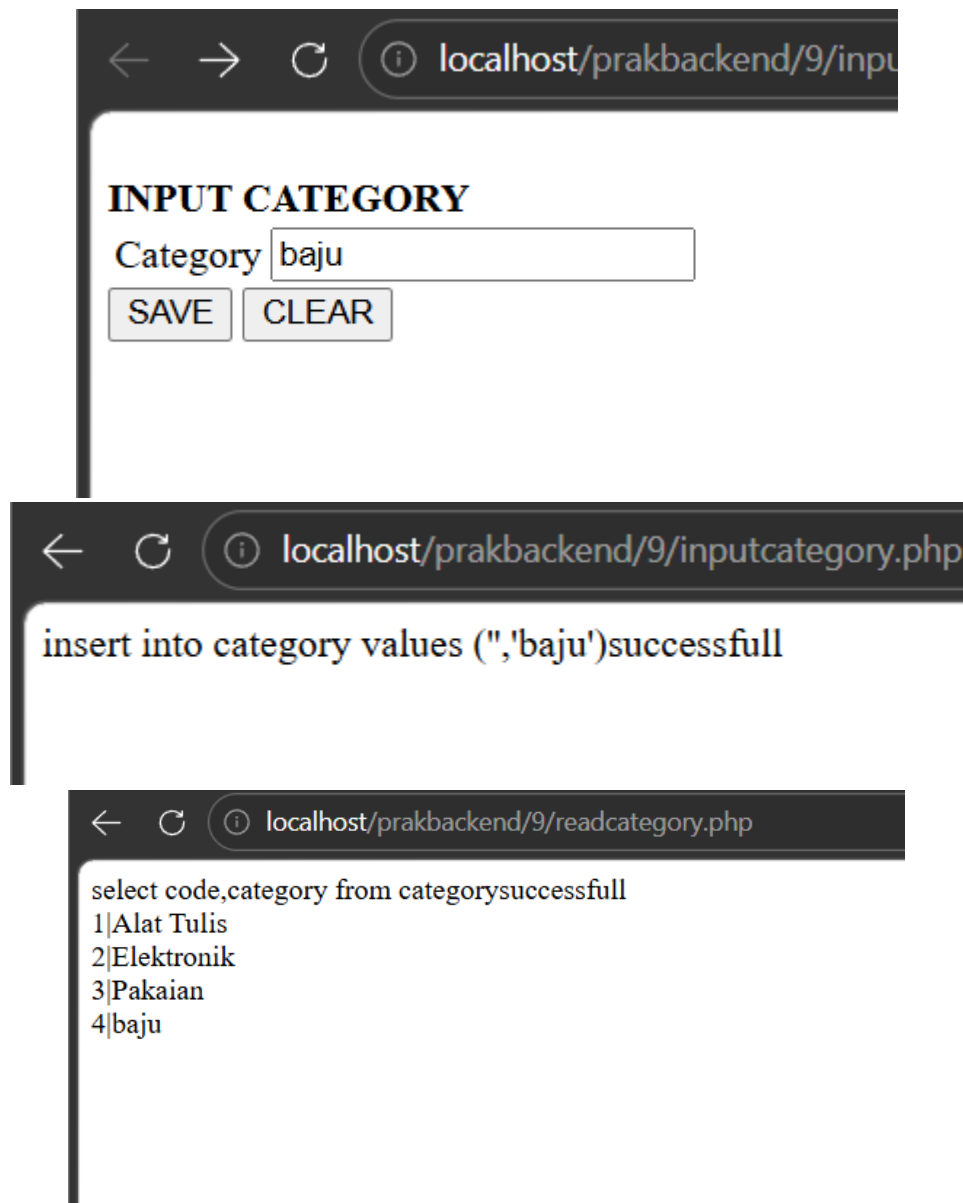
MariaDB [storedb]> SELECT code, category FROM category;
+----+-----+
| code | category |
+----+-----+
| 1    | Alat Tulis |
| 2    | Elektronik |
| 3    | Pakaian   |
+----+-----+
3 rows in set (0.001 sec)

MariaDB [storedb]>

```

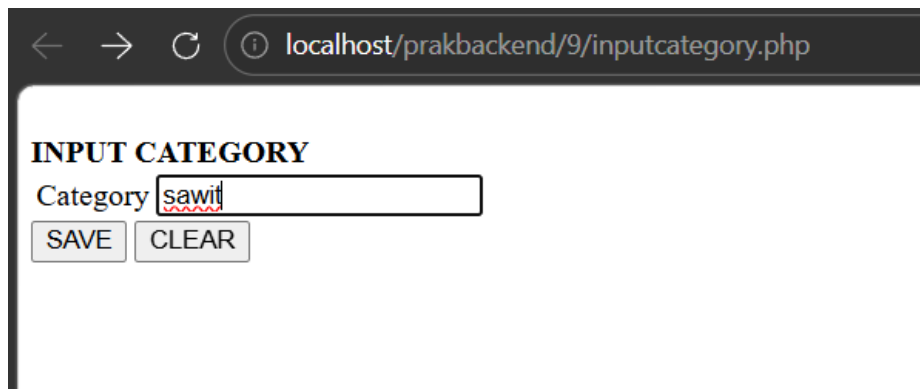
Pembahasan : Program praktik 2 adalah perintah SQL untuk membuat tabel category dan products beserta data awalnya. Tabel category memiliki dua kolom yaitu code sebagai primary key dengan AUTO_INCREMENT dan category bertipe VARCHAR(25). Tabel products memiliki tujuh kolom yaitu code, name, description, price, discount, status_code, dan category_code. Data awal dimasukkan menggunakan INSERT INTO dengan menyebutkan nama kolom secara eksplisit tanpa kolom code agar AUTO_INCREMENT mengisinya otomatis.

Praktik 3



Pembahasan : Program praktik 3 adalah program PHP yang membuat form input untuk mengisi tabel category dengan koneksi database langsung di dalam satu file. Koneksi database dilakukan menggunakan `mysqli_connect("localhost", "store", "store")` untuk terhubung ke server lalu `mysqli_select_db($id, "storedb")` untuk memilih database. Program menggunakan `isset($_POST['save_button'])` untuk memeriksa apakah form sudah disubmit, jika TRUE maka variabel `$StrSQL` diisi query `INSERT INTO category` menggunakan nilai `$_POST['category']` dari input form lalu dieksekusi dengan `mysqli_query($id, $StrSQL)`. Jika query berhasil maka `printf("successfull")` dicetak ke browser, jika gagal maka `printf("error")`. Jika form belum disubmit maka blok else menampilkan form HTML ke browser.

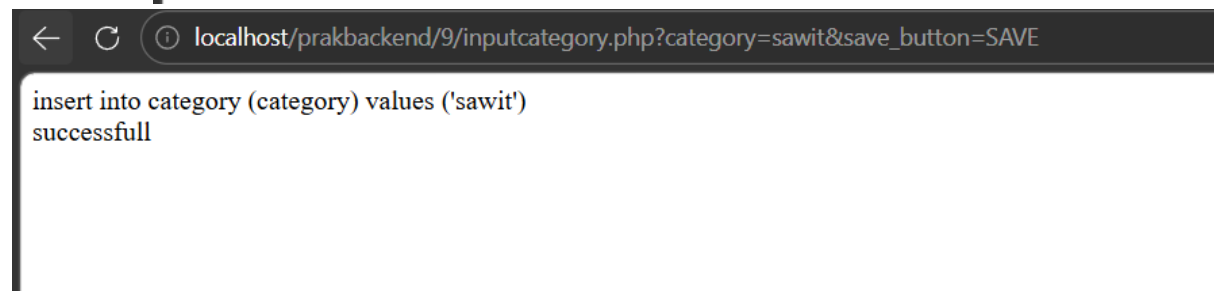
Praktik 4



← → ↻ ⓘ localhost/prakbackend/9/inputcategory.php

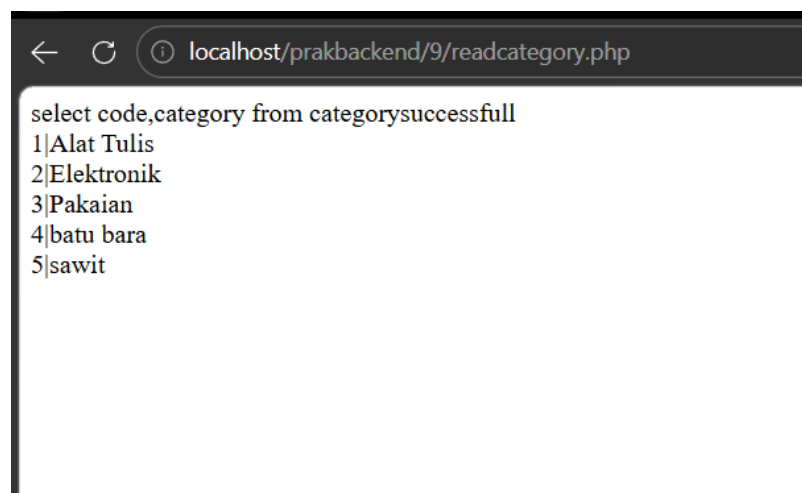
INPUT CATEGORY

Category



← ↻ ⓘ localhost/prakbackend/9/inputcategory.php?category=sawit&save_button=SAVE

insert into category (category) values ('sawit')
successfull



← ↻ ⓘ localhost/prakbackend/9/readcategory.php

select code,category from categorysuccessfull

- 1|Alat Tulis
- 2|Elektronik
- 3|Pakaian
- 4|batu bara
- 5|sawit

```

connect.php X inputcategory.php X
inputcategory.php
1 <html>
2 <head>
3 <title>input category</title>
4 </head><body bgcolor="#FFFFFF">
5 <?php
6 include 'connect.php';
7 if(isset($_POST['save_button']))
8 { $StrSQL="insert into category values ('','' . $_POST['category'] . '')";
9 echo $StrSQL;
10 $result=mysqli_query($id, $StrSQL);
11 if($result)
12 printf("successfull<br>");
13 else
14 printf("error");
15 }
16 else
17 {
18 }
19 <br>
20 <b>INPUT CATEGORY</b>
21 <form method="POST" name="forminputcategory">
22 <table border="0">
23 <tr><td>Category</td>
24 <td><input type="text" size="20" name="category"></td></tr>
25 </table>
26 <input type="submit" name="save_button" value="SAVE">
27 <input type="reset" value="CLEAR">
28 </form>
29 <?php
30 }
31 mysqli_close($id)
32 ?>
33 </body>
34 </html>

```

```

connect.php X
connect.php
1 <?php
2 $id = mysqli_connect("localhost", "root", "");
3 mysqli_select_db($id, "storedb");
4 ?>
...

```

Pembahasan : Program praktik 4 adalah versi kedua dari praktik 3 yang memisahkan koneksi database ke file tersendiri yaitu connect.php. File connect.php hanya berisi perintah mysqli_connect dan mysqli_select_db sehingga bisa digunakan ulang oleh file lain. File praktik_4.php menggunakan include 'connect.php' di bagian atas untuk mengambil koneksi dari file tersebut, sehingga variabel \$id hasil koneksi di connect.php langsung tersedia dan bisa digunakan untuk mysqli_query(\$id, \$StrSQL). Selain penggunaan include ini seluruh logika program identik dengan praktik 3.

Praktik 5

```
localhost/prakbackend/9/readcategory.php

select code,category from categorysuccessfull
1|Alat Tulis
2|Elektronik
3|Pakaian
4|batu bara
5|sawit
```

```
connect.php  readcategory.php X

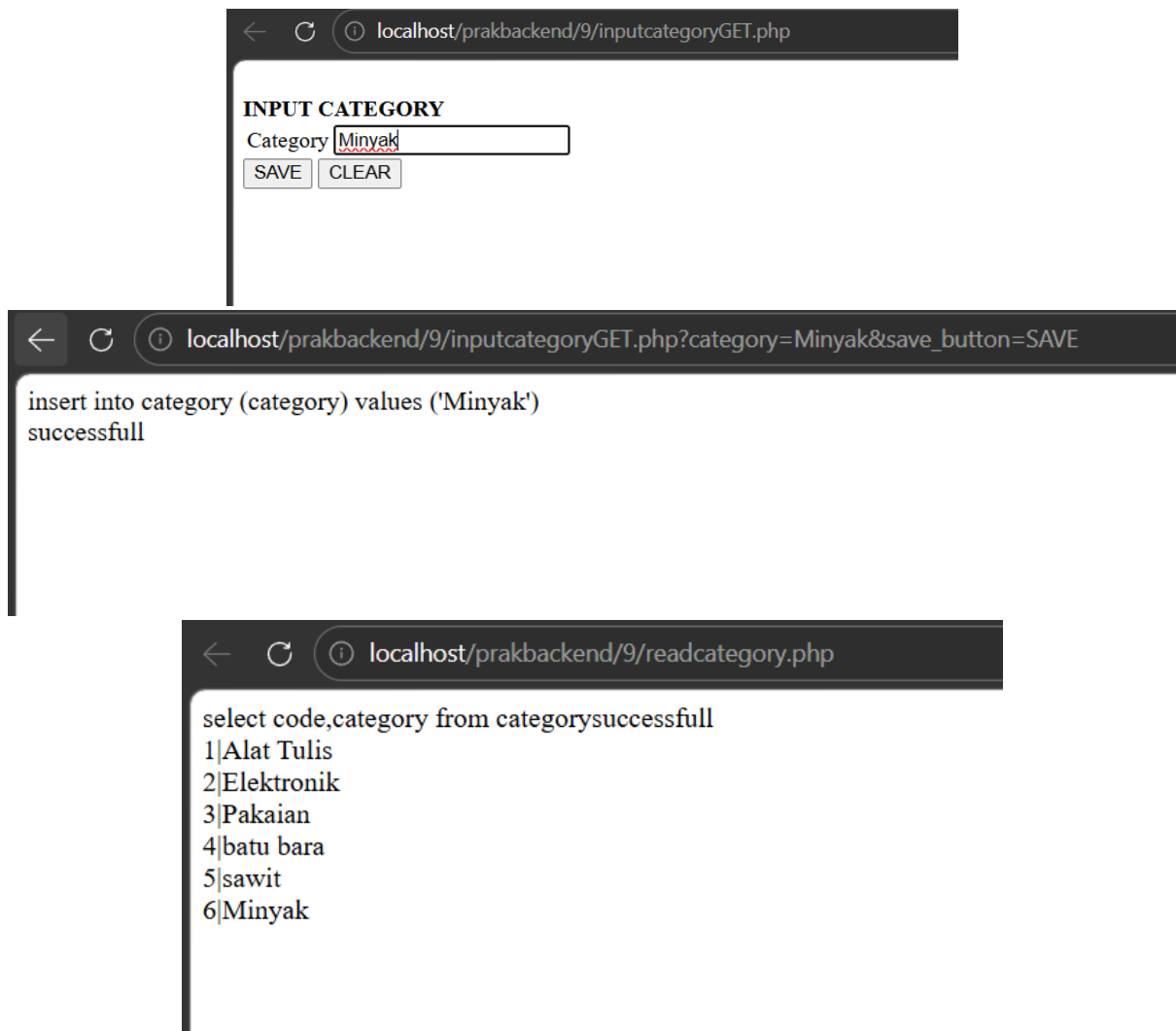
readcategory.php
1  <html>
2  <head>
3  <title>input category</title>
4  </head>
5  <body bgcolor="#FFFFFF">
6  <?php
7  include 'connect.php';
8  $StrSQL = "select code,category from category";
9  echo $StrSQL;
10 $result = mysqli_query($id, $StrSQL);
11 if ($result)
12     printf("successfull<br>");
13 else
14     printf("error");
15 while ($row = mysqli_fetch_row($result)) {
16     echo $row[0] . "|" . $row[1] . "<br>";
17 }
18 mysqli_close($id);
19 ?>
20 </body>
21 </html>
```


Pembahasan : rogram praktik 5 adalah program PHP untuk membaca dan menampilkan isi tabel category. Program menggunakan include 'connect.php' untuk koneksi database kemudian variabel \$StrSQL diisi query SELECT code, category FROM category lalu dieksekusi dengan mysqli_query(\$id, \$StrSQL). Hasil query ditampung di variabel \$result kemudian diiterasi menggunakan while (\$row = mysqli_fetch_row(\$result)) yang mengambil satu baris data setiap iterasinya ke variabel \$row, dimana \$row[0] berisi nilai kolom code dan \$row[1] berisi nilai kolom category yang dicetak ke browser dipisahkan karakter |.

LATIHAN

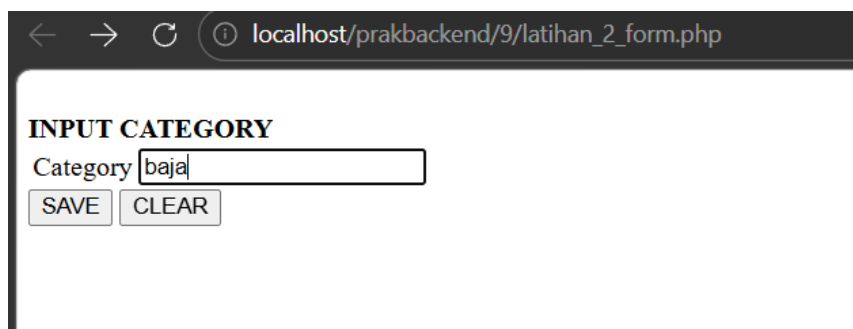
Latihan 1

```
connect.php  inputcategory.php X
inputcategory.php
1  <html>
2  <head>
3  <title>input category</title>
4  </head>
5  <body bgcolor="#FFFFFF">
6  <?php
7  include 'connect.php';
8  if (isset($_GET['save_button'])) {
9      $StrSQL = "insert into category (category) values ('" . $_GET['category'] . "')";
10     echo $StrSQL . "<br>";
11     $result = mysqli_query($id, $StrSQL);
12     if ($result)
13         printf("successfull<br>");
14     else
15         printf("error");
16 } else {
17     ?>
18     <br>
19     <b>INPUT CATEGORY</b>
20     <form method="GET" name="forminputcategory">
21         <table border="0">
22             <tr><td>Category</td>
23             <td><input type="text" size="20" name="category"></td></tr>
24         </table>
25         <input type="submit" name="save_button" value="SAVE">
26         <input type="reset" value="CLEAR">
27     </form>
28     <?php
29     }
30     mysqli_close($id);
31     ?>
32 </body>
33 </html>
```



Pembahasan : Latihan 1 mengubah program input category menjadi menggunakan metode GET. Perubahannya terletak pada atribut `method="GET"` di tag form dan penggunaan `$_GET['save_button']` serta `$_GET['category']` sebagai pengganti `$_POST`, sehingga data yang dikirim akan terlihat di URL browser.

Latihan 2



← ↻ localhost/prakbackend/9/latihan_2_action.php

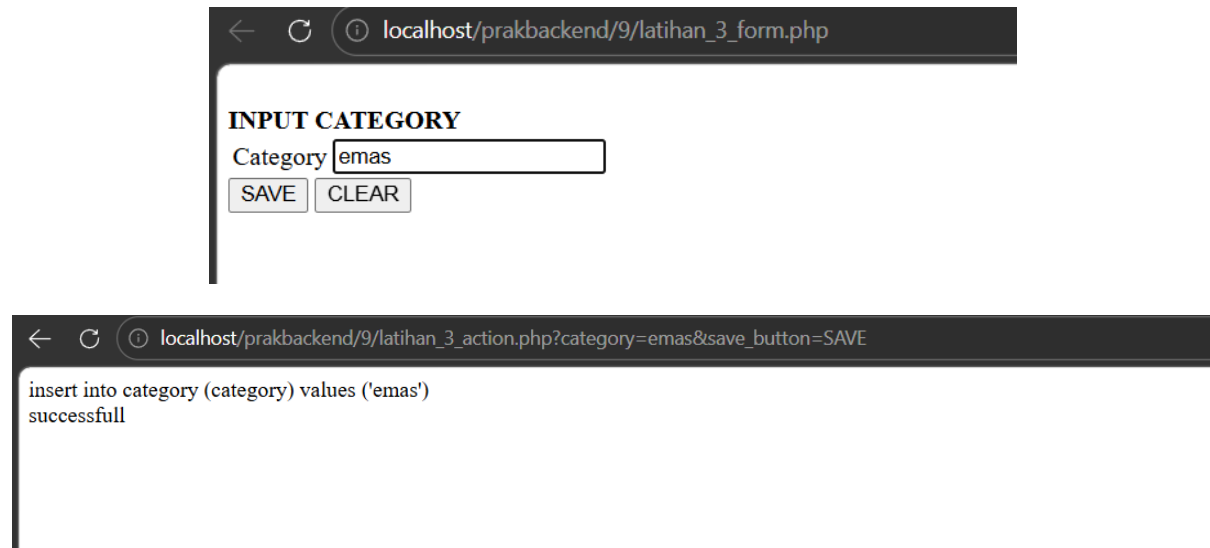
insert into category (category) values ('baja')
successfull

```
connect.php × inputcategory.php inputcategoryGET.php latihan_2_action.php ×
latihan_2_action.php
1 <html>
2 <body>
3 <?php
4 include 'connect.php';
5 $StrSQL = "insert into category (category) values ('" . $_POST['category'] . "')";
6 echo $StrSQL . "<br>";
7 $result = mysqli_query($id, $StrSQL);
8 if ($result)
9     printf("successfull<br>");
10 else
11     printf("error");
12 mysqli_close($id);
13 ?>
14 </body>
15 </html>
16
```

```
connect.php inputcategory.php inputcategoryGET.php latihan_2_form.php ×
latihan_2_form.php
1 <html>
2 <head>
3 <title>input category</title>
4 </head>
5 <body bgcolor="#FFFFFF">
6 <br>
7 <b>INPUT CATEGORY</b>
8 <form method="POST" action="latihan_2_action.php" name="forminputcategory">
9     <table border="0">
10         <tr><td>Category</td>
11         <td><input type="text" size="20" name="category"></td></tr>
12     </table>
13     <input type="submit" name="save_button" value="SAVE">
14     <input type="reset" value="CLEAR">
15 </form>
16 </body>
17 </html>
18
```

Pembahasan : Latihan 2 memisahkan program input category metode POST menjadi dua file terpisah. File latihan_2_form.php hanya berisi form HTML dengan method="POST" dan action="latihan_2_action.php", sedangkan file latihan_2_action.php berisi logika koneksi database dan eksekusi query INSERT menggunakan \$_POST['category']. Form harus dibuka terlebih dahulu melalui latihan_2_form.php sebelum data bisa dikirim ke latihan_2_action.php.

Latihan 3



The image shows two browser screenshots. The top screenshot displays a form titled "INPUT CATEGORY" with a text input field containing "emas" and two buttons labeled "SAVE" and "CLEAR". The bottom screenshot shows the result of a POST request to "latihan_3_action.php", displaying the SQL query "insert into category (category) values ('emas')" and the status "successfull".

localhost/prakbackend/9/latihan_3_form.php

INPUT CATEGORY

Category

localhost/prakbackend/9/latihan_3_action.php?category=emas&save_button=SAVE

insert into category (category) values ('emas')

successfull

```
connect.php inputcategory.php inputcategoryGET.php latihan_3_action.php X
latihan_3_action.php
1 <html>
2 <body>
3 <?php
4 include 'connect.php';
5 $StrSQL = "insert into category (category) values ('" . $_GET['category'] . "')";
6 echo $StrSQL . "<br>";
7 $result = mysqli_query($id, $StrSQL);
8 if ($result)
9     printf("successfull<br>");
10 else
11     printf("error");
12 mysqli_close($id);
13 ?>
14 </body>
15 </html>
16

connect.php X inputcategory.php inputcategoryGET.php latihan_3_form.php X
latihan_3_form.php
1 <html>
2 <head>
3 <title>input category</title>
4 </head>
5 <body bgcolor="#FFFFFF">
6 <br>
7 <b>INPUT CATEGORY</b>
8 <form method="GET" action="latihan_3_action.php" name="forminputcategory">
9     <table border="0">
10         <tr><td>Category</td>
11         <td><input type="text" size="20" name="category"></td></tr>
12     </table>
13     <input type="submit" name="save_button" value="SAVE">
14     <input type="reset" value="CLEAR">
15 </form>
16 </body>
17 </html>
18
```

Pembahasan : Latihan 3 memisahkan program input category metode GET menjadi dua file terpisah. Strukturnya sama dengan latihan 2 namun menggunakan method="GET" pada form dan \$_GET['category'] pada file action, sehingga data category terlihat pada URL saat form disubmit.

Latihan 4

```
connect.php inputcategory.php inputcategoryGET.php latihan_4.php X
latihan_4.php
1 <html>
2 <head>
3 <title>read category</title>
4 </head>
5 <body bgcolor="#FFFFFF">
6 <?php
7 include 'connect.php';
8 $StrSQL = "select code,category from category";
9 echo $StrSQL . "<br>";
10 $result = mysqli_query($id, $StrSQL);
11 if ($result)
12 | printf("successfull<br>");
13 else
14 | printf("error");
15 ?>
16 <table border="1" cellpadding="8" cellspacing="0">
17 | <tr>
18 | | <th>Code</th>
19 | | <th>Category</th>
20 | </tr>
21 | <?php
22 | while ($row = mysqli_fetch_row($result)) {
23 | | echo "<tr>";
24 | | echo "<td>" . $row[0] . "</td>";
25 | | echo "<td>" . $row[1] . "</td>";
26 | | echo "</tr>";
27 | }
28 | ?>
29 </table>
30 <?php
31 mysqli_close($id);
32 ?>
33 </body>
34 </html>
35
```

Pembahasa : Latihan 4 mengubah tampilan program readcategory dari teks biasa menjadi tabel HTML. Perbedaannya terletak pada penambahan tag <table>, <tr>, dan <td> di dalam loop while (\$row = mysqli_fetch_row(\$result)) sehingga setiap baris data ditampilkan sebagai baris tabel dengan kolom code dan category yang rapi di browser.

C. PEMBAHASAN TUGAS

D. KESIMPULAN

Dari praktikum ini dapat disimpulkan bahwa untuk mengakses database MySQL melalui PHP diperlukan tiga langkah utama yaitu koneksi ke server menggunakan `mysqli_connect`, pemilihan database menggunakan `mysqli_select_db`, dan eksekusi query menggunakan `mysqli_query`. Penggunaan file `connect.php` yang di-include memudahkan pengelolaan koneksi karena tidak perlu menulis ulang kode koneksi di setiap file. Data dari form HTML dapat dikirim menggunakan metode POST maupun GET dan diakses menggunakan superglobal `$_POST` atau `$_GET`, kemudian dimasukkan ke dalam query SQL untuk dieksekusi ke database. Hasil query SELECT dapat diiterasi menggunakan `mysqli_fetch_row` untuk menampilkan data satu baris per iterasi ke browser.

E. LAMPIRAN